METRIC

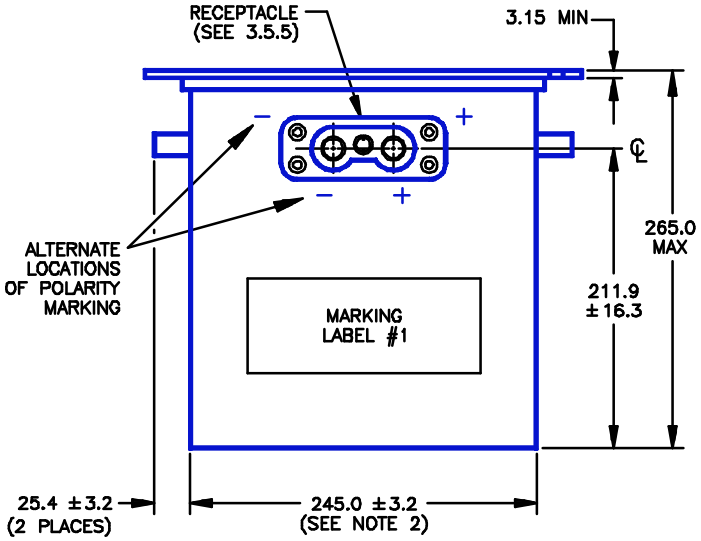
MIL-B-8565/15
07 October 1994

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1,
MAINTENANCE-FREE, 24-VOLT, 35-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-8565



FRONT VIEW

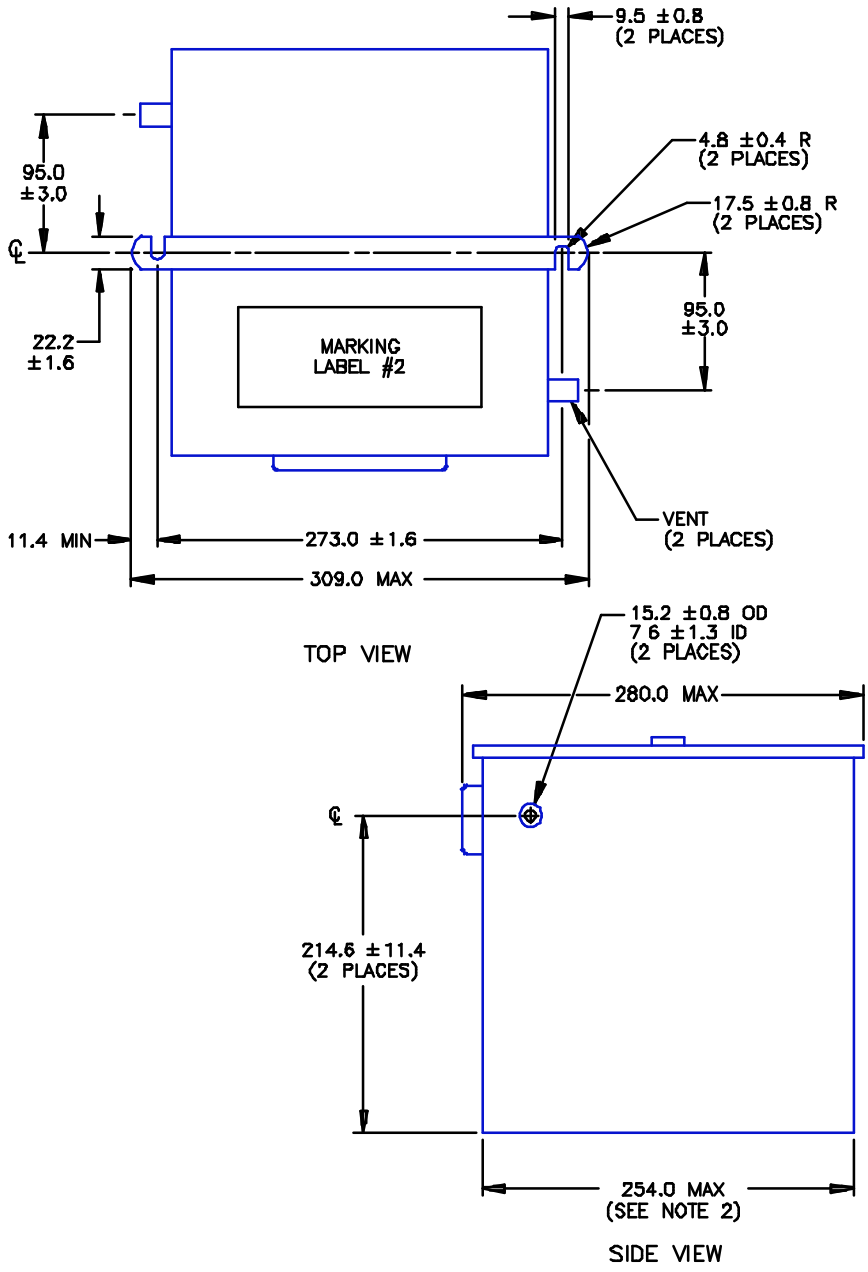
NOTES:
1. All dimensions are in millimeters.
2. Allow additional 4 mm width for cover.
3. Abbreviations used in this figure are maximum (MAX), minimum (MIN) and part number (P/N).

FIGURE 1. PN D8565/15-1 dimensions.

AMSC N/A FSC 6140

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NOTES:

1. All dimensions are in millimeters.
2. Allow additional 4 mm length for cover.
3. Abbreviations used in this figure are inside diameter (ID), outside diameter (OD), maximum (MAX), minimum (MIN), part number (PN) and radius (R).

FIGURE 1. PN D8565/15-1 dimensions-Continued.

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REQUIREMENTS:

1. The dimensions of the battery shall be as shown on figure 1.

2. The part number of the battery shall be D8565/15-1.

3. The rated capacity of the battery shall be 35 ampere-hours, based on its ability to conform to the life cycle final discharge (35.0 Ah/1 Hr/24°C/20.0 V).

4. The weight of the battery shall not exceed 40.82 kilograms (90.0 pounds).

5. Table I lists the quality assurance tests.

6. Delete all requirements for threaded-post connected batteries.

7. The battery shall conform to MIL-B-8565 for type 1, medium rate, except as modified by the following.

3.5.5 Receptacles. Add the following: "For D8565/15-1, the female receptacle (front view) shall be dimensionally equal to MS3509, style 1. The receptacle may be integrally molded in the container."

3.6.3.2.3 Battery caution marking (type 1 battery only). Delete and substitute the following: "Marking for Label 2 shown on figure 1 shall show the following with lettering height not less than 0.25 inch.

MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

PROCESS THIS BATTERY IN ACCORDANCE WITH
APPROVED PROCEDURES ONLY

SEE NAVAIR 17-15BAD-1
OR APPLICABLE SERVICE T.O."

3.6.8 Capacity and electrical performance. Delete "18.0 volts" and substitute "20.0 volts". Add the following: "The battery shall comply with the following requirements during the 14.0-volt constant voltage discharge:

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Requirement	24°C	-26°C
a. Minimum Current at Peak Value (Time = 0^{+})	1300 amperes	800 amperes
b. Minimum Current after 15 seconds	1000 amperes	450 amperes
c. Minimum Capacity delivered after 60 seconds	13.67 ampere-hours	5.93 ampere-hours

Inability to deliver the specified minimum discharge current and capacity shall constitute a test failure."

3.6.10 Life. Delete and substitute the following: "Batteries delivered under this specification shall be capable of at least 4 years of service life, demonstrated by successfully completing the cycling test of 4.6.13.1."

3.6.10.1 Life cycling (type 1 battery). Delete "eighty-five (85)" and substitute "one hundred thirty-five (135)".

4.5.1.3 Constant-potential charging. Delete and substitute the following: "Unless otherwise specified, the battery shall be charged at a constant potential of 28.5 ± 0.1 volts for 4.0 ± 0.1 hours with a power source having a current limit of 25 ± 1 amperes."

4.6.9 Constant current conditioning charge. Delete "18.0V" and substitute "20.0V".

4.6.10 Capacity discharge. Delete "18.0 volts" and substitute "20.0 volts".

4.6.13.1 Cycling test (type 1 battery). Delete "85 cycles" and substitute "135 cycles". Delete "2-hour charge" and substitute "4-hour charge".

4.6.14 Capacity discharge at 49°C (120°F). Delete "18.0 volts" and substitute "20.0 volts".

4.6.15 Charge and discharge test at low temperature. Delete "18.0V" and substitute "20.0 volts".

4.6.18 Mechanical shock. Delete and substitute the following:

"4.6.18.1 Functional shock. The battery shall be subjected to the functional shock test, procedure I, method 516.4 of MIL-STD-810. The test shall consist of 15 impacts, three in each direction of each orthogonal axis, except that which would place the battery in an inverted position. The shock pulse shall be in accordance with figure 516.4-4 using amplitude (a) and time duration (c). The battery shall be discharged at the C/2 rate during the test and, after each shock, shall be checked for any failure. No resilient mounting shall be provided for this test. At the conclusion of the test, the battery shall meet the specified requirements of 3.6.11 and 3.6.12.

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"4.6.18.2 Crash hazard. The battery shall be subjected to the crash hazard test, procedure V, method 516.4 of MIL-STD-810. The test shall consist of 10 impact shocks, two in each direction along each orthogonal axis, except that which would place the battery in an inverted position. The shock pulse shall be in accordance with figure 516.4-4 using amplitude (a) and time duration (c). No resilient mounting shall be provided for this test. The battery shall be inspected after each shock for any abnormalities. There shall be no cracks in or breakage of the battery mounting attachments, and the battery assembly shall remain in place and not create a hazard. However, mechanical bending or distortion shall be permitted, and the battery shall not be required to meet any electrical requirements at the conclusion of the test."

4.6.21 Temperature rise and float. Delete "18.0 volts" and substitute "20.0 volts".

4.6.23 Vibration. Delete and substitute the following: "The battery shall be subjected to the random vibration test for equipment installed in jet aircraft, category 5, procedure I, method 514.4 of MIL-STD-810. The battery shall be vibrated along each orthogonal axis using the vibration spectrum per figure 514.4-8 of MIL-STD-810, with W_O equal to 0.07 g^2/Hz. Test time per axis shall be 1 hour. The battery shall be charged in accordance with paragraph 4.5.1.3 during vibration and shall be discharged at the 6.5C rate for 1 minute prior to beginning each axis. The cables connected to the battery shall consist of four size AN-6 cables (two connected to the positive terminal and two connected to the negative terminal) and shall be clamped to the test apparatus at a length of 46 ± 3 centimeters from the battery. The battery shall be examined for physical damage resulting from the vibration exposure and shall meet the requirements of paragraph 3.6.16."

4.6.26 Ground storage. Delete "18.0 volts" and substitute "20.0 volts".

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TABLE I. Inspection of batteries and order of test.

Order of test	Description of inspection	Requirements paragraph	Inspection method paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample number			Group A tests	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.5, 3.6.1, 3.6.3 & 3.6.22	4.6.3	X	X	X	X	X	X
2	Dimensions & weight	3.6.5	4.6.4	X	X	X	X	X	
3	Dielectric strength	3.6.4	4.6.5	X	X	X	X	X	X
4	Strength of vent tubes	3.6.7	4.6.7	X	X	X	X	X	X
5	Color & marking	3.6.3	4.6.8	X	X	X	X	X	X
6	Conditioning charge	4.5.1.4	4.6.9	X	X	X		X	X
7	Capacity discharge	3.6.8	4.6.10	X	X	X	X	X	X
8	Constant voltage discharge	3.6.8	4.6.11	X	X		X	X	
9	Strength of receptacle	3.6.9	4.6.12			X			X
10	Life cycling	3.6.10	4.6.13	X				X	
11	Capacity discharge at 49°C	3.6.8	4.6.14		X				
12	Charge & discharge test at low temperature	3.6.8	4.6.15		X				X
13	Discharge while inverted	3.5.8 & 3.6.8	4.6.16		X				X
14	Altitude	3.6.11 & 3.6.12	4.6.17		X				X
15	Temperature shock	3.6.11 & 3.6.12	4.6.19	X				X	
16	Temperature rise & float	3.6.12 & 3.6.13	4.6.21		X				X
17	Battery gas emission test	3.6.11 & 3.6.14	4.6.22		X				
18	Vibration	3.6.11, 3.6.12 & 3.6.16	4.6.23		X				
19	Humidity	3.6.11, 3.6.12 & 3.6.17	4.6.24		X				X
20	Salt fog	3.6.11, 3.6.12 & 3.6.18	4.6.25	X				X	
21	Ground storage	3.6.15.1	4.6.26			X			
22	Physical integrity at high temperature	3.6.11 & 3.6.12	4.6.27	X					
23	Mechanical shock	3.6.11 & 3.6.12	4.6.18	X					
24	Impact resistance test	3.6.19	4.6.27.1		1/				
25	Final examination	3.6.4	4.6.5 & 4.6.30	X	X	X		X	X

1/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied. The containers shall be subjected to the impact resistance test of 4.6.27.

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